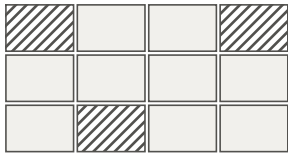


**a**

tiles of one section



 training tiles: fit the baseline and the probe

 held-out tiles: score both

baseline:  $\bar{v}_k(t)$ , the mean over cells of type  $t$

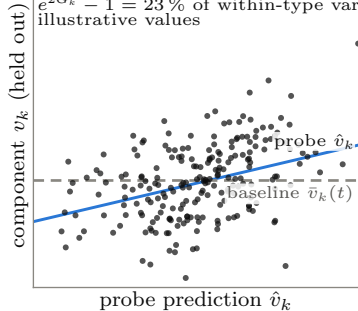
probe:  $\hat{v}_k(\mu_z, t)$ , ridge or MLP

**b**

$$G_k = \frac{1}{2} \log \frac{\text{MSE}_{\text{type}}}{\text{MSE}_{\text{probe}}} = 0.10 \text{ nats}$$

$$e^{2G_k} - 1 = 23\% \text{ of within-type variance}$$

illustrative values

**c**

 floor  this fit  uncontrolled

composition



fraction of uncontrolled = 0.31

image



fraction of uncontrolled = 0.16

$G_b$  (nats per component), illustrative